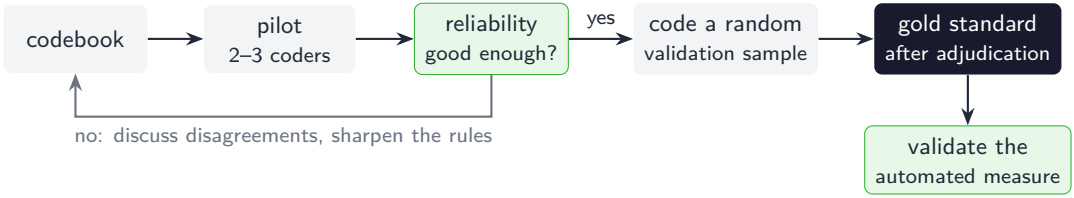
Bags, Vectors & Transformers

Annotation and Validation · Bonus, Part 5

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Validate, Validate, Validate

Every automated measure in this course (dictionary, classifier, BERT, LLM) is only as credible as its agreement with careful human judgment (Grimmer & Stewart, 2013).



The notebook simulates the coders, so it knows the truth and can show exactly what each measure does. In a real project, the coding is the expensive part; the code is the easy part.

The Codebook and the Sample

Running example: the dominant frame of a sentence

`economy`	costs, incomes, jobs, compensation
`environment`	nature, emissions, health effects
`politics`	parties, coalitions, votes
`human interest`	individual people and their stories

Decision rules are what coders need. "Code the frame of the main clause." "A protest is `politics` if it targets a decision, `human interest` if it's one person's story."

Drawing the sample

- **simple random sample** for validation: unbiased for the whole corpus
- rare categories: oversample them, then **weight**
- keep pilot and validation samples **separate**
- **blind** coders to predictions, sources and each other
- shuffle the order

Measuring Agreement

- **percent agreement**: share of items with the same code; rewards luck
- **Cohen's kappa**: corrects for chance, two coders

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

- **Krippendorff's alpha**: any number of coders, **missing codes**, nominal to interval data; the standard in communication science

$\alpha \geq .800$: reliable .667–.800: tentative conclusions only (Krippendorff, 2004)

Why percent agreement misleads

A rare code (5% of sentences). One coder reads carefully; the other always says "no" without reading.

percent agreement	**0.97**
Cohen's kappa	**0.00**

Always report a chance-corrected measure; for rare categories, also check agreement on the positive cases.

The Pilot: Who Is the Problem?

Three simulated coders, 200 sentences. A and B are careful; C has misread the codebook.

pair	agreement	kappa	alpha
A – B	0.86	0.82	0.82
A – C	0.81	0.74	0.73
B – C	0.82	0.76	0.75
A, B, C			0.77

Percent agreement with C still looks respectable; alpha makes the gap clearer.

One number says how much, not where. A confusion matrix between coders shows it: A–B disagree on scattered, mostly ambiguous sentences (noise). A–C show a **pattern**: C codes many environmental sentences as politics. That's a codebook or training problem: talk, add a rule, re-pilot.

Gold standard: two coders code everything, a third person settles disagreements. Report reliability *before* adjudication.

Validating a Classifier, Honestly

Report uncertainty

400 gold-standard sentences:

	estimate	95% CI
accuracy	0.81	0.77–0.85
F1, `human interest`	0.64	0.50–0.75

Wilson interval for accuracy; bootstrap for F1.

The rare category (40 sentences) is far less certain.

Validate per subgroup if you compare groups: a classifier can be fine overall and biased for one outlet, period or language.

How many documents?

Margin of error for an accuracy of 0.80:

documents	margin
100	±0.078
200	±0.055
400	±0.039
700	±0.030
1,600	±0.020

Halving the margin takes **four times** as many documents.

Proportions: When Accuracy Isn't Enough

Often the goal is a *share*: how much coverage uses the economy frame, per outlet. Counting the classifier's predictions is biased whenever its errors aren't balanced.

Correction for one category, from the validation set's true and false positive rates:

$$\hat{p} = \frac{\hat{p}_{\text{predicted}} - \text{FPR}}{\text{TPR} - \text{FPR}}$$

Economy frame, notebook data

	counted	truth
whole corpus	0.340	0.300
Agri Weekly	0.366	0.296
City Daily	0.322	0.302

Counting suggests the farming outlet uses the economy frame more (+4.4 points). In truth there's no difference. After per-outlet correction, the 95% interval for the difference includes zero.

LLMs as Annotators, and What to Report

LLMs can annotate well, sometimes better than crowd workers (Gilardi et al., 2023). But an LLM is one more coder to validate: **per task, dataset, prompt and model version**. Treat the prompt like a codebook: pilot, revise, freeze.

A useful benchmark: does the machine agree with humans about as well as humans agree with each other? In the notebook, adding the classifier as a "coder" drops alpha from 0.82 to 0.71.

Report

- the codebook and how it was developed
- sampling and sample size
- coders: who, trained how, blind to what
- reliability before adjudication (alpha, items, coders), per variable
- how disagreements were resolved
- validation of every automated measure, with confidence intervals, per subgroup if you compare groups

Tools: Label Studio, Doccano, INCEpTION (open source); Prodigy (paid); a spreadsheet for a few hundred items.

In the Notebook

- A codebook, a corpus, and drawing pilot and validation samples
- Simulated coders: percent agreement, kappa, Krippendorff's alpha, missing codes
- Confusion matrices between coders, and adjudication into a gold standard
- Validating a classifier with confidence intervals and per subgroup
- How many documents to annotate
- Biased proportions, a correction, and a spurious group difference

`bonus/notebooks/5_annotation_and_validation_exercises.ipynb` · 5 exercises · no GPU needed